WORKED SOLUTION

STEP III 2000, Question 13

Probability and Statistics • Pgf

STEP III 2000, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q13

The key idea

Both random variables are easiest to handle through "tail" or "cumulative" events. The first six comes after roll r exactly when *every* die misses on every one of the first r rolls; the last six has come by roll r exactly when *each* die has already shown a six. Each of these is an intersection of independent events, so its probability is a product, and the distribution follows by subtracting consecutive values. Once the first-six variable is recognised as geometric, everything about it is quick.

Solution

Throughout we assume the dice behave independently of each other and from roll to roll. For a single die, let X_i be the number of the roll on which die i first shows a six, so

$$\mathrm{P}(X_i > r) = q^r \qquad (r = 0, 1, 2, \dots),$$

since $X_i > r$ means r consecutive non-sixes. Then $R = \min_i X_i$ and $S = \max_i X_i$.

The first six

The event $R > r$ says that none of the n dice has shown a six in the first r rolls, that is, all nr individual throws were non-sixes. By independence,

$$\mathrm{P}(R > r) = q^{nr}.$$

Hence, for $r \geq 1$,

$$\mathrm{P}(R = r) = \mathrm{P}(R > r - 1) - \mathrm{P}(R > r) = q^{n(r-1)} - q^{nr} = q^{nr}(q^{-n} - 1),$$

which is the required result.

Writing $Q = q^n$, this is $\mathrm{P}(R = r) = Q^{r-1}(1 - Q)$: R is geometric, each roll of the whole set being a "success" (at least one six) with probability $1 - Q$. The probability

generating function is therefore a geometric series:

$$G_R(t) = \sum_{r=1}^{\infty} t^r Q^{r-1}(1-Q) = (1-Q)\,t \sum_{r=1}^{\infty} (Qt)^{r-1}$$
$$= \frac{(1-q^n)\,t}{1-q^n t},$$

valid for $|t| < q^{-n}$ (in particular near $t = 1$).

For the mean, differentiate by the quotient rule:

$$G_R'(t) = (1-q^n)\,\frac{(1-q^n t) + q^n t}{(1-q^n t)^2} = \frac{1-q^n}{(1-q^n t)^2},$$

so

$$\mathrm{E}(R) = G_R'(1) = \frac{1}{1-q^n}.$$

With $n = 2$ and $p = \frac{1}{6}$, $q^2 = \frac{25}{36}$, so $\mathrm{E}(R) = 1/\frac{11}{36} = \frac{36}{11}$, as required.

The last six

The event $S \le r$ says that every die has shown a six somewhere in the first r rolls. For one die this has probability $1 - q^r$, and the dice are independent, so

$$\mathrm{P}(S \le r) = (1-q^r)^n.$$

Subtracting consecutive values, for $r \ge 1$,

$$\mathrm{P}(S = r) = \mathrm{P}(S \le r) - \mathrm{P}(S \le r-1) = (1-q^r)^n - (1-q^{r-1})^n,$$

as required. (For $r = 1$ the second term is $0^n = 0$, correctly.)

> **Ask yourself** Summing $t^r[(1-q^r)^2 - (1-q^{r-1})^2]$ looks messy. How can we turn it into geometric series?

Expand the squares when $n = 2$:

$$\mathrm{P}(S = r) = (1 - 2q^r + q^{2r}) - (1 - 2q^{r-1} + q^{2r-2})$$
$$= 2q^{r-1}(1-q) - q^{2(r-1)}(1-q^2)$$
$$= 2p\,q^{r-1} - (1-q^2)\,(q^2)^{r-1}.$$

Each term is (a multiple of) a geometric distribution, so

$$G_S(t) = \sum_{r=1}^{\infty} t^r\,\mathrm{P}(S = r) = \frac{2pt}{1-qt} - \frac{(1-q^2)\,t}{1-q^2 t}.$$

This makes sense: since $S = X_1 + X_2 - R$, its distribution is "twice a geometric with parameter p, minus the distribution of R". To simplify, put everything over $(1-qt)(1-q^2t)$ and use $1 - q^2 = p(1+q)$:

$$2p(1-q^2t) - p(1+q)(1-qt) = p[\,2 - 2q^2t - 1 - q + qt + q^2t\,]$$
$$= p[(1-q) + qt(1-q)] = p^2(1+qt).$$

Hence

$$G_S(t) = \frac{p^2\, t\,(1+qt)}{(1-qt)(1-q^2t)}.$$

As a sanity check, $G_S(1) = \dfrac{p^2(1+q)}{p\cdot p(1+q)} = 1$.

For the mean, the split form is easiest to differentiate, since each piece is a geometric PGF whose derivative at 1 is its mean scaled by its coefficient. For $\dfrac{(1-a)t}{1-at}$ the derivative at $t=1$ is $\dfrac{1}{1-a}$ (exactly as for G_R above), so

$$\mathrm{E}(S) = G_S'(1) = 2\cdot\frac{1}{p} - \frac{1}{1-q^2} = \frac{2(1+q)-1}{p(1+q)} = \frac{1+2q}{p(1+q)}.$$

With $p=\frac{1}{6}$, $q=\frac{5}{6}$:

$$\mathrm{E}(S) = 12 - \frac{36}{11} = \frac{96}{11} \approx 8.73.$$

Checks

- $\mathrm{E}(R)+\mathrm{E}(S) = \frac{36}{11}+\frac{96}{11} = 12 = \mathrm{E}(X_1)+\mathrm{E}(X_2)$, as it must be since $\min+\max = X_1+X_2$.
- With $n=1$ the first and last six coincide, and G_R reduces to the single-die geometric PGF $pt/(1-qt)$.
- A Monte Carlo simulation of 2×10^5 pairs of dice gave sample means 3.274 and 8.709, against $\frac{36}{11}\approx 3.273$ and $\frac{96}{11}\approx 8.727$; summing the series for G_S numerically agrees with the closed form at several values of t.

Common slips

- Writing $\mathrm{P}(S=r)$ as "some die first shows a six on roll r and the others earlier", and forgetting the case where both dice first show a six on the same roll. Working with $\mathrm{P}(S\le r)$ avoids this entirely.
- Differentiating the simplified G_S instead of the split form: correct, but far more algebra and many chances for error.

Beyond STEP

R and S are the minimum and maximum of independent random variables, the simplest examples of *order statistics*; the trick of using $\mathrm{P}(\min > r)$ and $\mathrm{P}(\max \le r)$ works for any distribution. The memoryless geometric distribution is the discrete cousin of the exponential, and in the continuous limit the expected maximum of n exponentials becomes the harmonic sum $1+\frac{1}{2}+\cdots+\frac{1}{n}$ (times the mean), a key step in the coupon

collector's problem. Grimmett and Welsh, *Probability: An Introduction*, treats generating functions and these ideas in depth.

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